COMSATS UNIVERSITY ISLAMABAD, LAHORE CAMPUS

Department of Computer Engineering

MID TERM EXAMINATION — SPRING 2025 (SOLVED)

Course Title:	Machine Learning	**Course Code:**	CSC354 (Credit Hours: 3(3,0))
Course Instructor:	Dr. Ikramullah Khosa	**Program Name:**	BS Computer Engineering
Section / Date:	Section A, B | 15-04-2025	**Total Marks:**	25 Marks (Q1: 12, Q2: 6, Q3: 7)

Question 1 (PLO1-CLO1-C3) (12 Marks)

Relate the fundamental concepts of machine learning and answer the following short questions.

Question 1(A) **(1 Mark)**

Describe the purpose of regularization.

Solution:

The primary purpose of **regularization** is to **prevent overfitting (high variance)** and improve a machine learning model's ability to **generalize** accurately to new, unseen test examples. Regularization keeps all input features in the model while adding a penalty term (such as $\frac{\lambda}{2m}\sum_{j=1}^{n}\theta_j^2$) to the cost function $J(\theta)$ in order to **shrink the magnitude (values) of the parameters** $\theta_1, \theta_2, \ldots, \theta_n$ **toward zero**, producing a simpler, smoother hypothesis curve.

Question 1(B) **(1 Mark)**

What kind of cost function is expected in the case of an artificial neural network classifier: convex or non-convex?

Solution:

An artificial neural network (ANN) classifier with hidden layers has a **non-convex** cost function $J(\Theta)$. Because a multi-layer neural network composes non-linear activation functions (such as the sigmoid function) across multiple layers, its cost surface contains multiple **local minima (local optima)** and saddle points rather than a single bowl-shaped global minimum. *(Note: While a single-neuron logistic regression classifier has a convex log-loss cost function, a general multi-layer ANN cost function is non-convex.)*

Question 1(C) **(2 Marks)**

State one advantage and one disadvantage of using polynomial features with linear regression.

Solution:

- **One Advantage (1 Mark):** Adding polynomial features (such as x^2, x^3, x_1x_2) enables a linear regression model to capture **non-linear relationships and curved trends** in the data (reduc-

ing underfitting / high bias) while still allowing parameters to be learned using linear regression optimization techniques.

- **One Disadvantage (1 Mark):** High-degree polynomial features easily cause **overfitting (high variance)** (where the curve oscillates wildly and fails to generalize to new data) and trigger a **combinatorial feature explosion** ($\mathcal{O}(n^2)$ for quadratic terms and $\mathcal{O}(n^3)$ for cubic terms), making computation prohibitively expensive when the number of original features n is large.

Question 1(D) **(2 Marks)**

An artificial neural network classifier is designed to classify digits (0–9), using image as input with a size 10×10 pixels. State the number of neurons in the input layer and the output layer of the network.

Solution:

1. **Number of Neurons in the Input Layer (1 Mark):** Each input image has dimensions 10×10 pixels. Unrolling the 2D pixel grid into an input feature vector $x = [x_1, x_2, \ldots, x_n]^T$ gives:

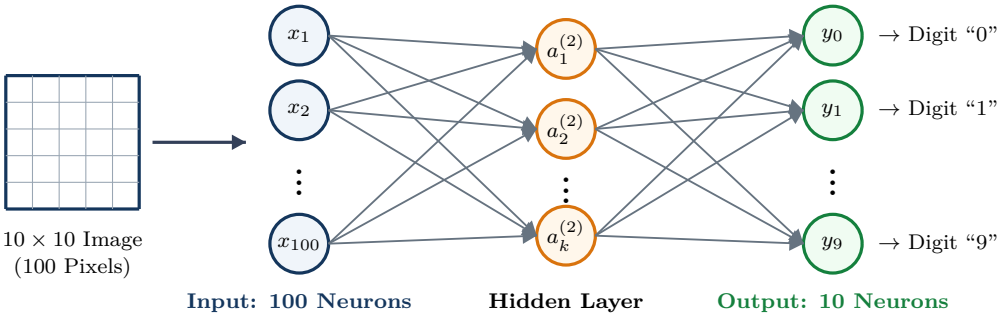
$$s_1 = n = 10 \times 10 = \mathbf{100} \text{ input neurons} \quad (\text{or } \mathbf{101} \text{ units if including the bias unit } x_0 = 1)$$

(Note: For standard grayscale digit images there are 100 *input neurons; if RGB color channels were used, there would be* $10 \times 10 \times 3 = 300$ *input neurons.)*

2. **Number of Neurons in the Output Layer (1 Mark):** Since there are 10 digit classes $(0, 1, 2, 3, 4, 5, 6, 7, 8, 9)$, the output layer uses One-vs-All / one-hot encoding ($h_\Theta(x) \in \mathbb{R}^{10}$) with one neuron per class:

$$s_{\text{output}} = K = \mathbf{10} \text{ output neurons} \quad (\text{representing digits 0 through 9})$$

Architecture Diagram: 10×10 **Digit Classifier Neural Network**



Question 1(E) **(3 Marks)**

For a Logistic Regression problem with binary classification ($y = 0/1$), the hypothesis is as follows:

$$h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

Assume that the optimized parameters are obtained using some unknown classifier as follows:

$$\theta = \begin{bmatrix} -4 \\ -2 \\ -2 \end{bmatrix}$$

Write the mathematical expression for the decision boundary. Plot the decision boundary for binary classification on feature space. Label the positive and negative class regions.

Solution:

1. **Mathematical Expression for the Decision Boundary (1 Mark):** Substituting $\theta_0 = -4$, $\theta_1 = -2$, and $\theta_2 = -2$ into the linear argument $z = \theta^T x$ gives:

   $$z = \theta_0 + \theta_1 x_1 + \theta_2 x_2 = -4 - 2x_1 - 2x_2$$

   The **Decision Boundary** is defined where $h_\theta(x) = g(z) = 0.5 \iff z = 0$:

   $$\boxed{-4 - 2x_1 - 2x_2 = 0 \quad \iff \quad x_1 + x_2 = -2 \quad (\text{or } x_2 = -x_1 - 2)}$$

   Axis Intercepts for Plotting:

   - x_1**-intercept** ($x_2 = 0$)**:** $-4 - 2x_1 = 0 \implies x_1 = -2 \implies (\mathbf{-2}, \mathbf{0})$.
   - x_2**-intercept** ($x_1 = 0$)**:** $-4 - 2x_2 = 0 \implies x_2 = -2 \implies (\mathbf{0}, \mathbf{-2})$.

2. **Positive Class ($y = 1$) and Negative Class ($y = 0$) Regions (1 Mark):**

   - **Positive Class Region ($y = 1$):** Predicted when $h_\theta(x) \geq 0.5 \iff \theta^T x \geq 0$:

     $$-4 - 2x_1 - 2x_2 \geq 0 \quad \iff \quad -2(x_1 + x_2) \geq 4 \quad \iff \quad \mathbf{x_1 + x_2 \leq -2}$$

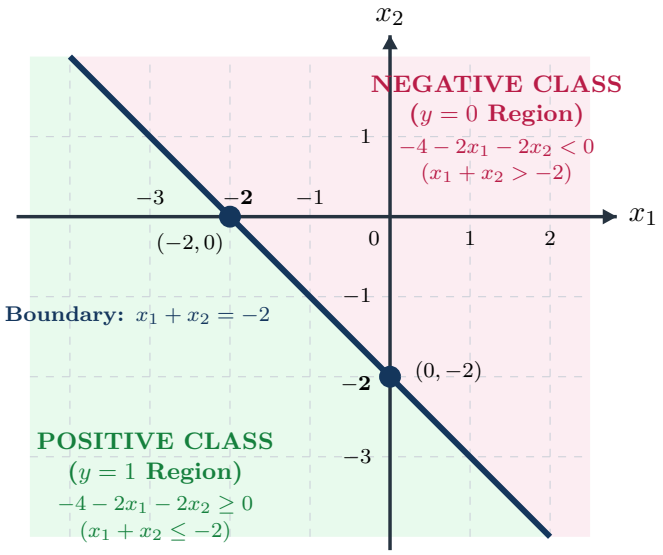
     *(Note: Dividing by -2 reverses the inequality sign from $\geq$ to $\leq$! For example, at test point $(-3, -3)$, $z = -4 - 2(-3) - 2(-3) = +8 \geq 0 \implies y = 1$. Thus, the region **below and to the left** of the line $x_1 + x_2 = -2$ is the **Positive Class** ($y = 1$) region.)*

   - **Negative Class Region ($y = 0$):** Predicted when $h_\theta(x) < 0.5 \iff \theta^T x < 0$:

     $$-4 - 2x_1 - 2x_2 < 0 \quad \iff \quad \mathbf{x_1 + x_2 > -2}$$

     *(At the origin $(0, 0)$, $z = -4 - 0 - 0 = -4 < 0 \implies y = 0$. Thus, the region **above and to the right** of the line $x_1 + x_2 = -2$ is the **Negative Class** ($y = 0$) region.)*

3. **Feature Space Plot of the Decision Boundary (1 Mark):**



Question 1(F) (3 Marks)

Show a perceptron model (with sigmoid activation function) performing as NAND gate for binary input (add a bias as third input). Estimate and assign appropriate weights with the inputs. Show the truth table with actual NAND gate output and the perceptron output.

Solution:

1. **Weight Estimation for the `NAND` Perceptron (1 Mark):** A digital `NAND` (NOT-AND) gate outputs 1 when at least one input is 0, and outputs 0 **only** when both $x_1 = 1$ and $x_2 = 1$. Using the sigmoid activation function $g(z) = \frac{1}{1+e^{-z}}$ (where $g(z) \approx 1$ for $z \geq +10$ and $g(z) \approx 0$ for $z \leq -10$), we assign:

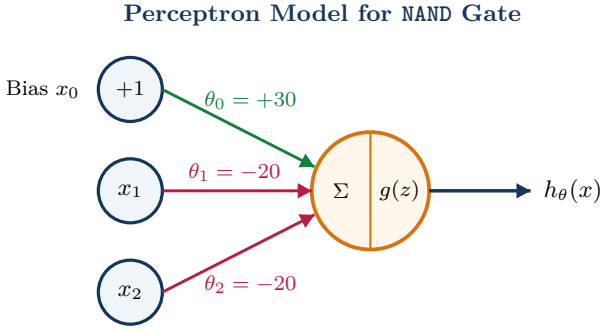
   $\theta_0 = +30$ (Bias weight for $x_0 = 1$), $\theta_1 = -20$ (Weight for x_1), $\theta_2 = -20$ (Weight for x_2)

   Thus, our sigmoid perceptron model is:

   $$h_\theta(x) = g\left(30 - 20x_1 - 20x_2\right) = \frac{1}{1 + e^{-(30-20x_1-20x_2)}}$$

   (Note: Any proportional weights satisfying $\theta_0 + \theta_1 > 0$, $\theta_0 + \theta_2 > 0$, and $\theta_0 + \theta_1 + \theta_2 < 0$—such as $[+15, -10, -10]$ or $[+3, -2, -2]$—also correctly implement the `NAND` gate.)

2. **Perceptron Model Diagram & Truth Table (2 Marks):**



x_1	x_2	$z = 30 - 20x_1 - 20x_2$	$h_\theta(x) = g(z)$	**Actual `NAND`**
0	0	$30 - 0 - 0 = +30$	$g(30) \approx \mathbf{1}$	**1**
0	1	$30 - 0 - 20 = +10$	$g(10) \approx \mathbf{1}$	**1**
1	0	$30 - 20 - 0 = +10$	$g(10) \approx \mathbf{1}$	**1**
1	1	$30 - 20 - 20 = -10$	$g(-10) \approx \mathbf{0}$	**0**

Question 2 (PLO1-CLO1-C3) (6 Marks)

Question 2 Statement **(6 Marks)**

Consider the following data to train a linear regression classifier: where X = Three examples as input (one in each row), Y = corresponding outputs, Θ_1–Θ_3 = three model hypotheses. Compute the sum of absolute error $\sum |X\Theta - Y|$ for each hypothesis and identify the best hypothesis model.

$$X = \begin{bmatrix} 1 & 10 \\ 1 & 21 \\ 1 & 14 \end{bmatrix}, \quad Y = \begin{bmatrix} 200 \\ 270 \\ 223 \end{bmatrix}, \quad \Theta_1 = \begin{bmatrix} 150 \\ 6 \end{bmatrix}, \quad \Theta_2 = \begin{bmatrix} 195 \\ 1.5 \end{bmatrix}, \quad \Theta_3 = \begin{bmatrix} 230 \\ 0.8 \end{bmatrix}$$

Solution:

We compute the predicted output vector $\hat{Y}_k = X\Theta_k$, the absolute error vector $|X\Theta_k - Y|$, and the Sum of Absolute Error $\text{SAE}(\Theta_k) = \sum_{i=1}^{3} |(X\Theta_k)_i - Y_i|$ for each of the three hypotheses:

1. Evaluation of Hypothesis Model 1: $\Theta_1 = \begin{bmatrix} 150 \\ 6 \end{bmatrix}$

$$X\Theta_1 = \begin{bmatrix} 1 & 10 \\ 1 & 21 \\ 1 & 14 \end{bmatrix} \begin{bmatrix} 150 \\ 6 \end{bmatrix} = \begin{bmatrix} 1(150) + 10(6) \\ 1(150) + 21(6) \\ 1(150) + 14(6) \end{bmatrix} = \begin{bmatrix} 150 + 60 \\ 150 + 126 \\ 150 + 84 \end{bmatrix} = \begin{bmatrix} 210 \\ 276 \\ 234 \end{bmatrix}$$

$$|X\Theta_1 - Y| = \left| \begin{bmatrix} 210 \\ 276 \\ 234 \end{bmatrix} - \begin{bmatrix} 200 \\ 270 \\ 223 \end{bmatrix} \right| = \left| \begin{bmatrix} 10 \\ 6 \\ 11 \end{bmatrix} \right| = \begin{bmatrix} 10 \\ 6 \\ 11 \end{bmatrix} \implies \sum |X\Theta_1 - Y| = 10 + 6 + 11 = \mathbf{27}$$

2. Evaluation of Hypothesis Model 2: $\Theta_2 = \begin{bmatrix} 195 \\ 1.5 \end{bmatrix}$

$$X\Theta_2 = \begin{bmatrix} 1 & 10 \\ 1 & 21 \\ 1 & 14 \end{bmatrix} \begin{bmatrix} 195 \\ 1.5 \end{bmatrix} = \begin{bmatrix} 1(195) + 10(1.5) \\ 1(195) + 21(1.5) \\ 1(195) + 14(1.5) \end{bmatrix} = \begin{bmatrix} 195 + 15 \\ 195 + 31.5 \\ 195 + 21 \end{bmatrix} = \begin{bmatrix} 210 \\ 226.5 \\ 216 \end{bmatrix}$$

$$|X\Theta_2 - Y| = \left| \begin{bmatrix} 210 \\ 226.5 \\ 216 \end{bmatrix} - \begin{bmatrix} 200 \\ 270 \\ 223 \end{bmatrix} \right| = \left| \begin{bmatrix} 10 \\ -43.5 \\ -7 \end{bmatrix} \right| = \begin{bmatrix} 10 \\ 43.5 \\ 7 \end{bmatrix} \implies \sum |X\Theta_2 - Y| = 10 + 43.5 + 7 = \mathbf{60.5}$$

3. Evaluation of Hypothesis Model 3: $\Theta_3 = \begin{bmatrix} 230 \\ 0.8 \end{bmatrix}$

$$X\Theta_3 = \begin{bmatrix} 1 & 10 \\ 1 & 21 \\ 1 & 14 \end{bmatrix} \begin{bmatrix} 230 \\ 0.8 \end{bmatrix} = \begin{bmatrix} 1(230) + 10(0.8) \\ 1(230) + 21(0.8) \\ 1(230) + 14(0.8) \end{bmatrix} = \begin{bmatrix} 230 + 8 \\ 230 + 16.8 \\ 230 + 11.2 \end{bmatrix} = \begin{bmatrix} 238 \\ 246.8 \\ 241.2 \end{bmatrix}$$

$$|X\Theta_3 - Y| = \left|\begin{bmatrix}238\\246.8\\241.2\end{bmatrix} - \begin{bmatrix}200\\270\\223\end{bmatrix}\right| = \left|\begin{bmatrix}38\\-23.2\\18.2\end{bmatrix}\right| = \begin{bmatrix}38\\23.2\\18.2\end{bmatrix} \quad \Longrightarrow \quad \sum |X\Theta_3 - Y| = 38 + 23.2 + 18.2 = \mathbf{79.4}$$

4. Summary Table & Identification of Best Model

Hypothesis Model	Example 1 Error	Example 2 Error	Example 3 Error	Sum of Absolute Error $\sum \lvert X\Theta - Y\rvert$
$\boldsymbol{\Theta_1 = [150,\ 6]^T}$	$\lvert 210 - 200\rvert = 10$	$\lvert 276 - 270\rvert = 6$	$\lvert 234 - 223\rvert = 11$	**27.0 (Minimum Error — Best Model)**
$\Theta_2 = [195,\ 1.5]^T$	$\lvert 210 - 200\rvert = 10$	$\lvert 226.5 - 270\rvert = 43.5$	$\lvert 216 - 223\rvert = 7$	60.5
$\Theta_3 = [230,\ 0.8]^T$	$\lvert 238 - 200\rvert = 38$	$\lvert 246.8 - 270\rvert = 23.2$	$\lvert 241.2 - 223\rvert = 18.2$	79.4

Final Conclusion for Question 2

Since $\sum |X\Theta_1 - Y| = \mathbf{27}$ is the smallest sum of absolute error among the three candidates ($27 < 60.5 < 79.4$), **Hypothesis Model** $\Theta_1 = \begin{bmatrix}150\\6\end{bmatrix}$ **is the best hypothesis model**.

Question 3 (PLO2-CLO2-C4) (7 Marks)

Question 3 Statement **(7 Marks)**

Illustrate the computational graph for the following (sigmoid) function:

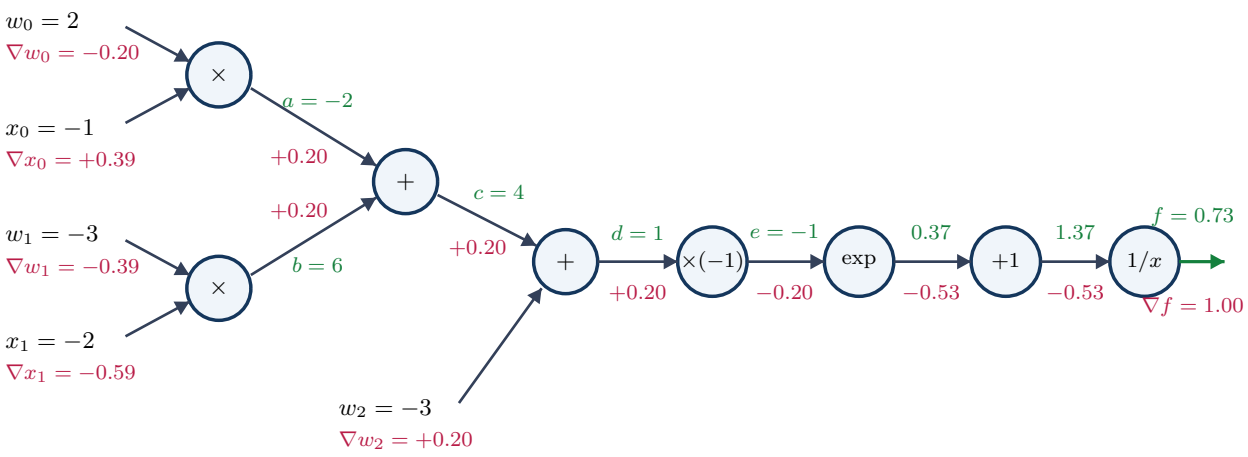
$$f(w,x) = \frac{1}{1+e^{-(w_0x_0+w_1x_1+w_2)}}$$

Perform the forward pass and backward pass. Compute and show the scores during the forward pass and the gradients during the backward pass. Use the following values for the variables:

$$w_0 = 2, \qquad x_0 = -1, \qquad w_1 = -3, \qquad x_1 = -2, \qquad w_2 = -3$$

Solution:

1. Complete Computational Graph (Forward Scores in Green Above, Backward Gradients in Red Below)



2. Step-by-Step Forward Pass (Computing Intermediate Scores)

We break $f(w,x) = \frac{1}{1+e^{-(w_0x_0+w_1x_1+w_2)}}$ into 8 elementary gates from left to right:

1. **Multiply Gate 1 (a):** $a = w_0 \cdot x_0 = (2)(-1) = \mathbf{-2}$
2. **Multiply Gate 2 (b):** $b = w_1 \cdot x_1 = (-3)(-2) = \mathbf{+6}$
3. **Add Gate 1 (c):** $c = a + b = -2 + 6 = \mathbf{+4}$
4. **Add Gate 2 (d):** $d = c + w_2 = 4 + (-3) = \mathbf{+1}$ *(Linear logit $z = w_0x_0 + w_1x_1 + w_2 = 1$)*
5. **Negation Gate (e):** $e = (-1) \cdot d = (-1)(1) = \mathbf{-1}$
6. **Exponential Gate (u):** $u = e^e = e^{-1} \approx \mathbf{0.3679}$ $(\approx \mathbf{0.37})$
7. **Add-1 Gate (v):** $v = 1 + u = 1 + 0.3679 = \mathbf{1.3679}$ $(\approx \mathbf{1.37})$
8. **Reciprocal Gate (f):** $f(w,x) = \frac{1}{v} = \frac{1}{1.3679} \approx \mathbf{0.7311}$ $(\approx \mathbf{0.73})$

3. Step-by-Step Backward Pass (Computing Gradients via the Chain Rule)

Starting from the output with base gradient $\frac{\partial f}{\partial f} = \mathbf{1.0000}$, we backpropagate right-to-left using (upstream gradient) × (local gradient):

1. **Backprop through Reciprocal Gate $f = \frac{1}{v}$ (local derivative $\frac{\partial f}{\partial v} = -\frac{1}{v^2}$):**

$$\frac{\partial f}{\partial v} = \left(-\frac{1}{v^2}\right) \cdot \frac{\partial f}{\partial f} = -\frac{1}{(1.3679)^2} \cdot (1.0) = -\frac{1}{1.8711} \approx \mathbf{-0.5344} \quad (\approx \mathbf{-0.53})$$

2. **Backprop through $+1$ Gate $v = 1 + u$ (local derivative $\frac{\partial v}{\partial u} = 1$):**

$$\frac{\partial f}{\partial u} = (1) \cdot \frac{\partial f}{\partial v} = (1)(-0.5344) = \mathbf{-0.5344} \quad (\approx \mathbf{-0.53})$$

3. **Backprop through exp Gate $u = e^e$ (local derivative $\frac{\partial u}{\partial e} = e^e$):**

$$\frac{\partial f}{\partial e} = e^e \cdot \frac{\partial f}{\partial u} = \left(e^{-1}\right)(-0.5344) = (0.3679)(-0.5344) \approx \mathbf{-0.1966} \quad (\approx \mathbf{-0.20})$$

4. **Backprop through $\times(-1)$ Gate $e = -1 \cdot d$ (local derivative $\frac{\partial e}{\partial d} = -1$):**

$$\frac{\partial f}{\partial d} = (-1) \cdot \frac{\partial f}{\partial e} = (-1)(-0.1966) = \mathbf{+0.1966} \quad (\approx \mathbf{+0.20})$$

(Verification using the Sigmoid derivative shortcut: $\sigma'(d) = \sigma(d)(1 - \sigma(d)) = 0.7311(1 - 0.7311) = \mathbf{0.1966} \approx \mathbf{0.20}$.)

5. **Backprop through Add Gate $d = c + w_2$ (addition distributes the gradient equally with local derivative 1):**

$$\frac{\partial f}{\partial c} = (1) \cdot \frac{\partial f}{\partial d} = \mathbf{+0.1966}\ (\approx \mathbf{+0.20}), \qquad \boxed{\frac{\partial f}{\partial w_2} = (1) \cdot \frac{\partial f}{\partial d} = \mathbf{+0.1966}\ (\approx \mathbf{+0.20})}$$

6. **Backprop through Add Gate $c = a + b$ (gradient distributor):**

$$\frac{\partial f}{\partial a} = (1) \cdot \frac{\partial f}{\partial c} = \mathbf{+0.1966}\ (\approx \mathbf{+0.20}), \qquad \frac{\partial f}{\partial b} = (1) \cdot \frac{\partial f}{\partial c} = \mathbf{+0.1966}\ (\approx \mathbf{+0.20})$$

7. **Backprop through Multiply Gate $a = w_0 \cdot x_0$ (multiplication switches inputs):**

$$\boxed{\frac{\partial f}{\partial w_0} = x_0 \cdot \frac{\partial f}{\partial a} = (-1)(0.1966) = \mathbf{-0.1966}\ (\approx \mathbf{-0.20})}$$

$$\boxed{\frac{\partial f}{\partial x_0} = w_0 \cdot \frac{\partial f}{\partial a} = (2)(0.1966) = \mathbf{+0.3932}\ (\approx \mathbf{+0.39}\ \text{or}\ \mathbf{+0.40})}$$

8. **Backprop through Multiply Gate $b = w_1 \cdot x_1$ (multiplication switches inputs):**

$$\boxed{\frac{\partial f}{\partial w_1} = x_1 \cdot \frac{\partial f}{\partial b} = (-2)(0.1966) = \mathbf{-0.3932}\ (\approx \mathbf{-0.39}\ \text{or}\ \mathbf{-0.40})}$$

$$\boxed{\frac{\partial f}{\partial x_1} = w_1 \cdot \frac{\partial f}{\partial b} = (-3)(0.1966) = \mathbf{-0.5898}\ (\approx \mathbf{-0.59}\ \text{or}\ \mathbf{-0.60})}$$